What the ALMA Release of 6 November 2026 Can Test: Pre-Data Predictions for U4_28473 and U4_27928

Working note, v1.3 (2026-11-03; v1.0–v1.2 2026-11-01). AI-assisted research programme; not peer reviewed. $\kappa = \frac{1}{2}$ is fitted. The theory is not closed.

In one line. Expected: CO(3–2) is detected in U4_28473 and a rotation speed is scored. That speed will measure this galaxy's stellar and gas mass (its $\alpha_{\rm CO}$), not a_0: at 3 kpc the law adds about 10 km s^{-1} to Newton, while the mass inputs span hundreds. The sealed galaxy U4_27928 is 7.36″ off-axis and is expected to score nothing. One disc excludes no model. Each prediction, and what would count against it, is in Table 7. **v1.3** changes no prediction or number: it states route N's retirement, adds this line and the scorecard, and moves the v1.1 and v1.2 corrections to Section 8.

Abstract

Observing unit uid://A001/X3788/Xbfd2 of ALMA programme 2024.1.00100.L (CONDOR) is listed to become public at 02:42:31 UTC on 2026-11-06. We state this programme's predictions for it before any of its data can be read.

The unit's science target is U4_28473, at $z = 2.52$, not U4_27928, the galaxy this programme sealed for this date. U4_27928 lies 7.36″ off the phase centre, so its reading is serendipitous. At its empirical gas mass the frozen aperture gives S/N 0.16, so we expect that test to score nothing.

U4_28473 has no earlier seal, so we add one here. Its published CO(3–2) flux gives $\log M_{\rm gas} = 11.33$ ($\alpha_{\rm CO} = 4.36$) or 10.69 ($\alpha_{\rm CO} = 1$). At 3 kpc the baryons alone, under Newtonian gravity, reach 431–846 km s^{-1} across the input branches. The galaxy therefore tests its mass budget, not $a_0(z)$: where the CO reaches, g/a_0 is large and the readings differ by far less than the stellar-mass and $\alpha_{\rm CO}$ ranges. Real noise measured in a public cube of the same programme is 2.54 times below the archive's listed figure; with that noise and beam, mock discs of the published size are detected in 6 of 6 seeds, so a detection and a usable V_c are the expected outcome, though the unit's own noise is unknown until release. One disc excludes no model.

1 The framework, as it stands

The programme studies a galaxy law $g_N = g^2/(g + a_0)$ with $a_0 = \kappa c\sqrt{G\rho_{\rm DE}}$, on two footings never pooled ($a_0(0) = 9.3603 \times 10^{-11}$ and 1.1312×10^{-10} m s^{-2}), and a gravitating $w \simeq 0$ field component called cold dark energy, which is neither matter nor a particle. **Since v1.2 the cold sector's halo rule (route N, a core $S_{\rm cold} = f_c\,\eta(M_t)\,\Sigma_{\rm coll}(M_t)$ with no a_0 in it) has been retired** after failing on measured cluster gas; the programme's current statement is the law alone (PAPER67, DOI 10.5281/zenodo.23883741). Route N's rows below were computed and sealed before that and are kept for the record; none of them is a live prediction.

The record's current position is unfavourable and is stated here so the predictions are read in context. On measured gas no tested rule fits galaxies, clusters and lensing together with what has been derived. The law remains the best-performing galaxy description, with misses of its own (Milky Way K_z, IC 1613), and has no cluster fit. Rule N failed dwarfs, had a hand-chosen core, and is now retired. Nothing in this note changes that, and nothing here says the data favour the framework.

2 What is released

From the ALMA archive (ObsCore query, 2026-11-01): band 3, beam 0.066″, 22 680 s on source. Windows 94.79–96.67, 97.60–98.62, 106.75–108.73 and 108.74–110.61 GHz, with line sensitivity 0.42–0.45 mJy beam^{-1} per 10 km s^{-1}. The second window is centred on CO(3–2) at $z = 2.52$ (98.11 GHz). The dates and rights are the archive's and can change.

Correction to our own record. Our survey (CFG570) matched galaxies to archive rows by field of view. It listed U4_27928 as having CO(2–1) in this unit, and later files treated the unit as if aimed at it. KMOS$^{\rm 3D}$'s U4_28473 lies 0.002″ from the phase centre; U4_27928 lies 7.361″ away, at primary-beam response 0.96 for a 61″ beam. Its CO(2–1), at 95.29 GHz, does fall in the first window. The sealed test stands as written, as an off-axis reading. Whether the archive's imaged product extends to this offset at this resolution is unverified until release.

3 U4_27928: sealed requirements at the Hα speed

CFG571 (SHA-256 be27457d…) sealed the gas each model requires as a function of the measured V_c at 8.27 kpc. Addenda added $n = 1$, $n = 2$ and an AQUAL disc as labelled readings. None of these is altered here. Our Hα fit gives $v_{\rm rot} = 226.1$, $\sigma_0 = 77.8\,{\rm km\,s^{-1}}$, with a recorded median 28 % slow bias at the outermost radius. That gives V_c from 226 to 373 $\rm km\,s^{-1}$ depending on two branches never pooled, pressure support and bias (Table 1). The empirical scaling line used in the record gives $\log M_{\rm gas} = 10.51$.

Table 1: U4_27928: required total $\log M_{\rm gas}$ at the sealed $R = 8.27$ kpc, $R_g = R_d$, canonical / alt, at the Hα speed. FLOOR: the stars alone already exceed the requirement. Route N: lg M_{200} 12.5, added after the criteria, information only.

case	V_c	n=1 F-DESI	n=1 F-flat	$\nu_{\rm mono}$ F-DESI	n=2 F-DESI	n=1 R-H	route N
a P0	226	9.72 / 9.22	9.62 / 8.66	9.89 / 9.57	10.17 / 10.07	FLOOR / FLOOR	10.35
b P0 /0.72	314	10.91 / 10.88	10.90 / 10.87	10.94 / 10.92	10.97 / 10.96	10.72 / 10.66	11.01
c P-B	303	10.84 / 10.81	10.84 / 10.80	10.88 / 10.85	10.92 / 10.90	10.63 / 10.55	10.96
d P-B /0.72	373	11.17 / 11.15	11.17 / 11.15	11.19 / 11.18	11.21 / 11.20	11.06 / 11.02	11.23

Forecast. In the frozen $2R_e$ aperture over $\pm 500\,{\rm km\,s^{-1}}$, corrected for the primary beam, an extended disc reads S/N 0.16 at $\log M_{\rm gas} = 10.51$ and 0.49 at 11.0. The gate needs 5. The 3σ limit is $\log M_{\rm gas} = 11.79$ for $\alpha_{\rm CO} = 4.36$, above every framework requirement (at most 11.33), and 11.15 for $\alpha_{\rm CO} = 1$, below the requirement only in the fastest case. These numbers use the listed sensitivity; scaling by the measured ratios of Section 4 gives S/N about 0.6 at 10.51 (a hand estimate), still far below the gate. **Prediction A: G1 fails and no model is scored on this disc.** A much more compact or clumpy disc would change this.

4 U4_28473: new predictions

Inputs, never pooled. $z = 2.52455$ (KMOS$^{\rm 3D}$). CO(3–2) $0.79 \pm 0.08\,{\rm Jy\,km\,s^{-1}}$, gas R_e 1.56 kpc, stellar R_e 1.37 kpc at 4.4 μm (Tadaki et al. 2023, arXiv:2309.02703), with their $R_{31} = 1.8$. $\log M_\star$ 11.31 (Tadaki et al. 2017) and 11.35 (KMOS$^{\rm 3D}$), each also -0.2 dex. Stellar R_e 3.16 kpc as a second shape, from KMOS$^{\rm 3D}$'s rest-optical 0.383″. Distances: flat, $h = 0.6856$, $\Omega_m = 0.2976$, the programme's own background fit (CFG886), as frozen in the criteria. These public numbers were read before the predictions were made, so this is a forward prediction of unreleased kinematics, not a blind test of the mass inputs.

Readings. $n = 1$ (canonical, primary), $\nu_{\rm mono}$, $n = 2$, the v3 AQUAL disc, route N at lg M_{200} 12–13, and Newton. For $a_0(z)/a_0(0)$: flat 1, F-DESI 0.819, and the ends of our own BAO band, 0.325 and 1.212 (quoted at $z = 2.33$, extrapolated). Two rivals as sealed: $a_0 \propto H(z)$ (R-H, 3.805) and the feedback RAR-equivalent (L-fb, 4.88, held at its table's end).

Table 2: U4_28473: predicted V_c in $\rm km\,s^{-1}$ at $R = 2/3/4$ kpc from the published CO flux, canonical footing. S1: stellar R_e 1.37 kpc; S2: 3.16 kpc. Gas R_e 1.56 kpc.

lg $M_\star$, shape, $\alpha_{\rm CO}$	Newton	n=1 F-flat	n=1 F-DESI	n=1 R-H	route N 12.5
11.31, S1, 4.36	891/827/733	894/832/741	893/831/740	903/845/761	891/827/734
11.31, S1, 1	707/648/571	711/655/581	710/654/579	721/672/604	707/649/572
11.31, S2, 4.36	720/716/675	724/722/683	723/721/682	734/737/704	720/716/676
11.31, S2, 1	473/499/494	479/508/505	478/506/503	494/528/531	473/500/495
11.11, S1, 4.36	801/746/664	805/752/672	804/751/671	814/767/694	801/747/665
11.11, S1, 1	589/542/479	594/550/490	593/549/488	607/569/517	590/543/480
11.11, S2, 4.36	684/670/624	688/676/632	687/675/631	699/693/655	684/671/624
11.11, S2, 1	416/431/421	423/441/433	422/439/431	439/463/462	417/432/422

The AQUAL disc solution at lg $M_\star$ 11.31, S1 differs from the algebraic $n = 1$ in-plane value by a factor 0.990–1.007 at 1–8 kpc on the flat, F-DESI and R-H cells, so it adds nothing separable. Route N's core gives $V_{\rm core}(3\,{\rm kpc})$ about $25\,{\rm km\,s^{-1}}$ at lg M_{200} 12.5. Rule N has no derived redshift dependence; holding $\Sigma_{\rm coll}$ at its $z = 0$ value is an extrapolation, supported only by a halo acceleration ratio of 0.97–1.06 at $z = 2.33$ at fixed M_{200}.

Table 3: U4_28473: $\alpha_{\mathrm{CO,req}}$ at $R = 3\,\mathrm{kpc}$ for a measured V_c, $n = 1$, canonical. F = FLOOR, stars alone exceed the requirement. The primary statistic of the scoring plan.

V_c	lg $M_\star$ 11.31, S2			lg $M_\star$ 11.11, S2			lg $M_\star$ 11.11, S1		
	flat	F-DESI	R-H	flat	F-DESI	R-H	flat	F-DESI	R-H
350	F	F	F	0.08	0.10	F	F	F	F
400	F	F	F	0.56	0.58	0.32	F	F	F
450	0.30	0.32	0.04	1.10	1.12	0.85	F	F	F
500	0.90	0.92	0.64	1.71	1.73	1.44	0.33	0.35	0.07
550	1.57	1.59	1.30	2.38	2.40	2.11	1.00	1.02	0.73
600	2.31	2.33	2.03	3.11	3.13	2.84	1.73	1.75	1.46

Table 4: U4_28473: $V_c(3\,\mathrm{kpc})$ in $\mathrm{km\,s^{-1}}$ for one input branch (lg $M_\star$ 11.11, S2, $\alpha_{\mathrm{CO}} = 1$), every reading.

reading	canonical	alt
Newton	431	
$n = 1$, flat	441	443
$n = 1$, F-DESI	439	441
$n = 1$, band 0.325	435	435
$n = 1$, band 1.212	443	445
$n = 1$, R-H	463	469
$n = 1$, L-fb	471	477
ν_{mono}, flat	438	440
ν_{mono}, F-DESI	437	438
$n = 2$, flat	436	437
$n = 2$, F-DESI	435	436
route N, lg M_{200} 12.0	432	
route N, lg M_{200} 12.5	432	
route N, lg M_{200} 13.0	432	

Detection forecast at the listed sensitivity and resolution, smooth disc. Beam 0.54 kpc. Peak S/N 6.4 per beam per $100\,\mathrm{km\,s^{-1}}$; 3σ per beam out to 0.70 kpc; 3σ per azimuthal ring out to 2.21 kpc. Moment-0 aperture S/N over $\pm 500\,\mathrm{km\,s^{-1}}$ peaks at only 1.86, while a rotation-matched three-dimensional statistic reads 9.6. The 9.6 is an ideal matched filter with the true model known, not an achievable search.

Mock cubes at the listed noise (V_c $430\,\mathrm{km\,s^{-1}}$, published flux, 7 noise seeds each, so rates are rough; Table 5). Clumpy gas was not tested and could change these. The compact fits scatter low, and the rms recovery error is 8.9 %, larger than most quoted errors; one fit sits 4σ from truth. All four scored fits have a 90 % flux radius below 3 kpc, so they would be scored inside that radius.

Table 5: U4_28473 mock cubes at the listed noise. NOT ALMA data.

disc	moment-0 gate	G1-3D gate	fit scored
smooth, gas R_e 1.56 kpc	0 of 7	1 of 7	1 of 7
compact, gas R_e 0.8 kpc	0 of 7	6 of 7	4 of 7
no line	0 of 7	0 of 7	none

Real-cube checks. (i) The scoring code ran on the public cube of this galaxy from programme 2021.1.01291.S (beam $0.48'' \times 0.43''$): $1''$-aperture flux $0.92 \pm 0.12\,\mathrm{Jy\,km\,s^{-1}}$ against the published 0.79 ± 0.08, line redshift 2.52365 against 2.523 ± 0.004, and a velocity-shift mutation correctly fails. The disc fit there is not a usable speed, since a 3.9 kpc beam inflates σ_0, and it scores nothing. (ii) That cube's noise is $0.355\,\mathrm{mJy\,beam^{-1}}$ per $10\,\mathrm{km\,s^{-1}}$ against the listed 1.30. (iii) In CONDOR's public unit for UDS17159 the noise is 0.162 against the listed 0.411, a ratio of 2.54, from 30 channels at the window's edge; the beam is $0.21'' \times 0.16''$ against the listed $0.126''$.

Forecast with those ratios (Table 6; noise $0.171\,\mathrm{mJy\,beam^{-1}}$ per $10\,\mathrm{km\,s^{-1}}$, beam $0.109'' \times 0.085''$, about 0.9 kpc, so the inner curve is only partly resolved). Smooth discs reach a 90 % flux radius of 3.2–5.1 kpc. This

rests on one similar unit and smooth mocks; the release's own noise is measured when it is read.

Table 6: U4_28473 mock cubes with the measured noise and beam ratios, 6 seeds each, V_c 430 km s^{-1}. NOT ALMA data.

disc	moment-0 gate	G1-3D gate	fit scored	recovered V_c
smooth, gas R_e 1.56 kpc	6 of 6	6 of 6	6 of 6	408–444
compact, gas R_e 0.8 kpc	6 of 6	6 of 6	6 of 6	413–438
no line	0 of 6	0 of 6	none	–

Statements fixed before the data

1. **Prediction B0.** CO(3–2) is detected and a V_c is scored, if the noise and beam ratios of the similar unit hold. If the noise is as listed and the gas smooth, an upper limit is likely instead. Neither outcome bears on the law.
2. **Prediction B1 (every reading with $\nu \geq 1$, and route N).** $V_c(R)$ at or above the Newtonian value of the true baryons. At 3 kpc the lowest branch gives 431 km s^{-1} (lg $M_\star$ 11.11, S2, $\alpha_{\rm CO} = 1$). A lower V_c requires lower $M_\star$, a more extended stellar disc, $\alpha_{\rm CO} < 1$, or missing pressure support, and it is not a statement about a_0.
3. **Prediction B2.** If V_c(3 kpc) is near the 430 km s^{-1} of our low-S/N Hα fit, $\alpha_{\rm CO,req}$ is 0.88 for lg $M_\star$ 11.11, S2, 0.07 for 11.31, S2, and FLOOR for every S1 branch ($n = 1$, flat, canonical). $\alpha_{\rm CO} = 4.36$ with published masses is then out for every reading. The Hα fit is weak: flagged low-S/N, edge-limited, dispersion at the prior's bound.
4. **Prediction B3.** No pair among flat, F-DESI and the band will be separated. At the branch of Table 4, flat and F-DESI differ by 2 km s^{-1} at 3 kpc.
5. **Prediction B4.** R-H sits 22 km s^{-1} above flat at 3 kpc on the same branch, while the mass branches span hundreds. No separation from R-H is expected either, unless the masses are fixed independently.
6. **Prediction B5 (record only; route N retired).** The law exceeds route N by 9 km s^{-1} at 3 kpc and by 22 at 8 kpc on this branch. The forecast does not reach 8 kpc, so this release does not separate them.

Table 7: Scorecard for 6 November. Results are filled in only under the scoring rules of Section 6.

	Prediction	Counts against it	Result
A	U4_27928: G1 fails, nothing scored	G1 passes at its frozen aperture	
B0	U4_28473 detected, V_c scored	no detection at measured noise	
B1	V_c(3 kpc) $\geq$ Newton on the true baryons; lowest branch 431 km s^{-1}	V_c below 431 by more than its error: the mass inputs or pressure support are wrong, not a_0	
B2	At $V_c \approx 430$, $\alpha_{\rm CO} = 4.36$ with published masses is out for every reading	V_c(3 kpc) above 600, where Table 3 first nears 4.36	
B3	flat, F-DESI and the band not separated	any pair separated under the rule	
B4	R-H not separated	R-H separated	
B5	record only (route N retired)	–	

5 The other sealed test

GS4_24110 ([CI](2–1), programme 2025.1.01377.L, public 2026-11-28 by our earlier metadata) is unchanged (Table 8). A governing-plan analysis already found every $n = 1$ versus $n = 2$ pair at or below 0.51σ, and the forecast there at the listed sensitivity is also an upper limit, a forecast the noise ratios above may loosen. Other CONDOR targets have later dates (UDS16442 2026-11-25, ID2993 2026-12-04, GMASS1084 2026-12-22) and carry no predictions here.

Table 8: GS4_24110 (public 2026-11-28), sealed before this note, unchanged: required $\log M_{\rm gas}$ at R_e, V_c 206 km s^{-1}, $R_g = R_d$, canonical / alt.

lg $M_\star$	model	$\nu_{\rm mono}$	$n = 1$	v3 AQUAL disc
10.89	F-DESI	9.82 / 9.11	9.53 / FLOOR	9.96 / 9.66
10.89	F-flat	9.46 / FLOOR	FLOOR / FLOOR	9.77 / 9.10
10.89	R-H	FLOOR / FLOOR	FLOOR / FLOOR	FLOOR / FLOOR
10.76	F-DESI	10.43 / 10.33	10.37 / 10.27	10.46 / 10.39
10.76	F-flat	10.36 / 10.24	10.30 / 10.18	10.42 / 10.33
10.76	R-H	FLOOR / FLOOR	FLOOR / FLOOR	FLOOR / FLOOR

6 Scoring

Frozen in commit 57d3ac6b64, alone, before any script. **Detection** was amended before the data by addendum G1 (0113d554d3, owner-approved, chosen on mock cubes only). The frozen moment-0 gate is printed but no longer decides. The cube is smoothed to 0.15″ and 100 km s^{-1}; connected voxels $\geq 2.5\sigma$ with a peak $\geq 4\sigma$ touching a 0.5″ cylinder at the phase centre are summed, and the S/N against the same procedure at 200 off-source seeds must reach 5, with z within 0.004. A fit is scored only if no parameter sits at a bound, three starts agree within 15 %, and $\sigma(V_c)/V_c \leq 0.25$. The V_c error is the larger of the quoted one and the mock scatter. V_c from a thin-disc forward model, with inclination and pressure branches never pooled. The primary statistic is $\alpha_{\rm CO,req}$ at the measured V_c, with its interval from $V_c \pm 1\sigma$ and $M_\star \pm 0.2$ dex. A reading is *consistent* if that interval includes $\alpha_{\rm CO}$ in [0.5, 6], *in tension* if it lies wholly outside, and FLOOR if stars alone exceed the requirement even at $M_\star - 0.2$ dex. Two a_0 readings are separated only if their requirements differ by more than the $M_\star$ interval on both. Pooling, branch choice after the data, exclusion statements and law-form statements from one disc are forbidden. U4_27928 stays under its own plan (CFG806).

7 Limits

- One disc is a demonstration. No model is excluded.
- The detection gate was changed after mock tests and before the data. The first written expectation, that the moment-0 gate would pass, was wrong on mocks. Both are on record.
- The scoring code has run on one real pipeline product, at a much larger beam, and otherwise on mock cubes; it has not run on this unit. Its injection test starts at the true values and flatters recovery. Mock V_c errors understate the scatter in places.
- Stellar masses and fluxes keep the authors' reductions, including their cosmology for distances. Rivals and CFG571's angular size are as sealed.
- The rotation at these radii may not trace circular speed; pressure branches stay separate. An unmodelled bulge or AGN is not included.
- The E, E-CCBH and BH-k_3 a_0 branches are not recomputed at this redshift.
- Two rows were added after the criteria as information: route N on U4_27928 and $\alpha_{\rm CO,req}$ at 430 km s^{-1}. One script check failed first on a threshold not in the criteria; 82 cells matched exactly, and the correction is disclosed in the lane README.

8 Version history

v1.1 (2026-11-01). v1.0 quoted a rotation-matched S/N of 9.6 for U4_28473 without saying that it assumes the true disc model is known. A blind three-dimensional search on mock cubes at the archive noise detects the published smooth disc in 1 of 7 noise seeds. v1.1 added those rates, the detection gate filed before the data, and the scoring code.

v1.2 (2026-11-01). v1.1 said a non-detection of U4_28473 was likely, using the archive's listed sensitivity. A public cube from the same programme has real noise 2.54 times below its listed figure and a wider beam. With those ratios mock discs are detected in 6 of 6 seeds, so a detection and a usable V_c became the expected outcome, inferred from one similar unit.

v1.3 (2026-11-03). Route N's retirement stated; summary line, scorecard and this section added. In none of these versions did a prediction of V_c or $\alpha_{\rm CO,req}$ change.

9 Reproducibility

Lane `rebuild_2026_10_22/AC1_alma_condor_predictions`: criteria 57d3ac6b64, run 02ffa20dd8; all checks pass, and a mutation (ten times R_{31}) trips its check and exits 1. `ac1_results.json` SHA-256 bfdfed68.... Seals: CFG571 be27457d..., $n = 1$ 9802de0d..., $n = 2$ d64727b5..., AQUAL b5d65a5e.... Scoring code `ac1_score.py` (6874d71f1f), mock and validation outputs `ac1_score_power*`, `ac1_score_selftest.out`, `ac1_score_VALIDATION*`, noise check `ac1_condor_noise*` (7cdb58c138). An audit script ties every quoted number here to those files.

AI assistance: Anthropic's Claude (analysis, review, drafting). ALMA metadata: ALMA Science Archive. Sources: Tadaki et al. 2017, ApJ 834, 135; Tadaki et al. 2023, arXiv:2309.02703; Wisnioski et al. 2019 (KMOS$^{\rm 3D}$).

$\kappa = \frac{1}{2}$ fitted; footings never pooled; cold dark energy is required beyond galaxies and is not derived; not theory closed.